

Commercial Intake-to-ACORD Automation (v1 Deep Spec)

#Objective

Build a production-ready v1 system that converts commercial client intake into review-ready ACORD forms with high data quality, clear exception handling, and full auditability.

Initial form scope:

- ACORD 125 (Commercial Insurance Application)
- ACORD 126 (Commercial General Liability Section)
- ACORD 140 (Property Section)

#1) Product Scope and Non-Negotiables

##In Scope (v1)

- Intake capture (web form + document upload + optional email ingestion)
- Canonical data model for commercial submission data
- Mapping engine from canonical model to ACORD field set
- Prefill output package for 125/126/140
- Human QA/review workflow (Maria/team approval)
- Exception queue for missing/conflicting fields
- Export package + activity/audit logs

##Out of Scope (v1)

- Full carrier appetite recommendation engine
- Fully autonomous bind workflow
- E-sign/package automation beyond prefill outputs

##Non-Negotiables

- No form leaves review without human approval
- Required fields enforced before "Quote Ready"
- Every transformation is traceable (audit log)

#2) Required Canonical Data Model

#A) Account-Level (Insured)

- account_id
- legal_name
- dba_name
- fein
- years_in_business
- website
- naics_code
- operations_description
- contact_name

- contact_title
- email
- mobile
- mailing_address

#B) Location-Level

- location_id
- location_address
- occupancy_type
- construction_type
- year_built
- square_footage
- protection_class
- sprinklered (yes/no)
- burglar_alarm (yes/no)
- fire_alarm (yes/no)

#C) Exposure-Level

- annual_revenue
- payroll_total
- employee_count
- subcontractor_costs
- gross_sales_split (if needed)
- prior_carrier
- prior_premium
- requested_effective_date

#D) Loss/Claims

- claims_5yr_count
- claims_5yr_total_paid
- open_claims_count
- claim_detail_items[] (date, cause, paid/incurred)

#E) Coverage Intent

- lines_requested (GL, Property, etc.)
- limit_preferences
- deductible_preferences
- additional_insured_requirements
- certificate_requirements

#F) Producer/Workflow Meta

- lead_source
- owner_user
- intake_completed_at
- confidence_score
- missing_fields[]
- exception_flags[]

#3) Form Mapping Tables (v1)

#ACORD 125 Core Mapping (examples)

- Canonical 'legal_name' -> ACORD 125 Applicant Name
- 'mailing_address' -> Mailing Address block
- 'naics_code' + 'operations_description' -> Business Description / Classification
- 'years_in_business' -> Yrs in Business
- 'requested_effective_date' -> Proposed Effective Date
- 'prior_carrier' / 'prior_premium' -> Prior Coverage section

#ACORD 126 Core Mapping (examples)

- 'lines_requested' includes GL -> enable 126 section
- 'annual_revenue', 'payroll_total', 'employee_count' -> exposure basis fields
- 'claims_5yr_*' + 'claim_detail_items' -> Loss History fields
- 'subcontractor_costs' -> subcontractor exposure fields
- 'additional_insured_requirements' -> AI/COI section inputs

#ACORD 140 Core Mapping (examples)

- 'location_*' -> Location schedule blocks
- 'construction_type', 'occupancy_type', 'year_built', 'square_footage' -> property details
- 'sprinklered', 'fire_alarm', 'burglar_alarm' -> protection details
- deductible/limit preferences -> requested terms

Note: Maintain mapping table as versioned config (YAML/JSON), not hardcoded logic.

#4) Intake Requirements (Quote-Ready Gate)

Minimum required for Quote Ready:

- 1) Legal insured name
- 2) Primary contact + email/phone
- 3) Operations description + NAICS (or class code proxy)
- 4) Revenue + payroll
- 5) At least one location with address and occupancy/construction basics
- 6) Requested effective date
- 7) Prior carrier/premium (or explicit unknown flag)
- 8) Basic 5-year loss summary (or no-loss attestation)

Gate rule:

- If any required field missing -> status remains Engaged, create exception task.

#5) QA and Confidence Rules

#Confidence Scoring

- Start score at 1.0
- Deduct for each missing required field
- Deduct for contradictory values (e.g., payroll present, employee count = 0)
- Deduct for low-quality document extraction

Bands:

- High: ≥ 0.90 (fast review)
- Medium: $0.75 \leq 0.89$ (standard review)
- Low: < 0.75 (manual rebuild required)

#Validation Rules (examples)

- FEIN format check
- Effective date not in past unless renewal flow
- Revenue/payroll non-negative and reasonable ranges
- Location addresses valid format
- Claims totals align with claim items

#Review Checklist

- Field completeness
- Exposure consistency
- Coverage intent captured
- Loss narrative adequacy
- Producer notes added

#6) Exception Queue Design

Exception categories:

- Missing required data
- Conflicting values
- Mapping ambiguity
- Unsupported document format
- Carrier-specific requirement missing

Routing:

- CSR/Intake owner -> data completion
- Maria -> commercial underwriting quality review
- Producer -> final coverage strategy decisions

SLA:

- High-severity missing data: same business day
- Standard exceptions: within 24 hours

#7) Workflow States and Triggers

Lifecycle alignment:

- New -> Contact Attempted -> Engaged -> Quote Ready -> Proposal Sent -> Won/Lost

Automation triggers (v1):

- 1) Intake completed event -> run validation + mapping
- 2) If Quote Ready -> create ACORD package + assign review task
- 3) If exceptions -> create task + send missing-data request template
- 4) Review approved -> export package + notify producer queue

#8) Output Package Specification

For each submission, generate:

- Prefilled ACORD 125 PDF (or structured payload)
- Prefilled ACORD 126 PDF
- Prefilled ACORD 140 PDF
- Missing/assumption report
- Submission summary sheet (key exposures + notes)

Naming convention:

'{AccountName}-{YYYYMMDD}-ACORD-PACK-v{N}'

Storage:

- Canonical folder by account/year in Drive/CD document storage
- Immutable copy of original source docs

#9) Security, Compliance, and Audit

- Role-based access (intake vs reviewer vs producer)
- PII minimization in notifications
- Audit log entries for:
 - source ingest
 - field transforms
 - manual edits
 - approvals/exports
- Retention policy for source and generated artifacts

#10) Implementation Plan (Manus + Team)

#Phase 1 (Week 1-2)

- Finalize canonical schema
- Build 125 mapping + validator

- Stand up exception queue
- Pilot with 10 real submissions

#Phase 2 (Week 3-4)

- Add 126 + 140 mappings
- Improve extraction quality/rules
- Add review dashboard + confidence sorting

#Phase 3 (Week 5-6)

- Tighten workflow automation in CD/QuoteRUSH
- Add carrier-specific requirement templates
- Expand pilot volume and measure KPIs

#11) KPI Targets (First 60 Days)

- >=85% submissions reach review-ready in first pass
- <=24h median exception resolution time
- >=50% reduction in manual rekey time
- >=90% reviewer satisfaction on data completeness

#12) Immediate Next Actions

- 1) Confirm field dictionary with Maria for commercial quoting reality
- 2) Lock required vs optional fields list for Quote Ready gate
- 3) Build mapping config v0.1 (125 first)
- 4) Run first 10-file pilot and tune exceptions
- 5) Publish weekly quality scorecard